

NOTES

Definition

*If $f: X \rightarrow Y$ such that $f(x) = y$ is one-one and onto, then we can define a unique function $g: Y \rightarrow X$ such that $g(y) = x$. Then $g(y)$ is known as Inverse function of $f(x)$.

* **Trigonometric functions are not one-one functions But by restricting the domain inverse trigonometric functions can be defined.**

Domain & Range of Inverse Tri. Function

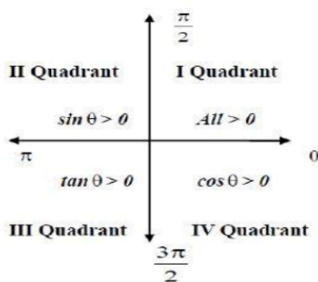
The value of an Inverse trigonometric function which lies in the range of Principal branch is called the Principal value of the Inverse trigonometric function.

Table For the Domain & Range (Principal Value Branch)

FUNCTION	DOMAIN	RANGE (PRINCIPAL VALUE BRANCH)
$\sin^{-1}$	$[-1, 1]$	$[-\frac{\pi}{2}, \frac{\pi}{2}]$
$\cos^{-1}$	$[-1, 1]$	$[0, \pi]$
$\tan^{-1}$	R	$(-\frac{\pi}{2}, \frac{\pi}{2})$
$\operatorname{cosec}^{-1}$	$R - (-1, 1)$	$[-\frac{\pi}{2}, \frac{\pi}{2}] - \{0\}$
$\sec^{-1}$	$R - (-1, 1)$	$[0, \pi] - \{\frac{\pi}{2}\}$
$\cot^{-1}$	R	$(0, \pi)$

Important Trigonometric Results

*All T-ratios changes in $\frac{\pi}{2} \pm \theta$ and $\frac{3\pi}{2} \pm \theta$ while remains same in $\pi \pm \theta$ and $2\pi \pm \theta$.



Important For Principal value of ITF (Inverse tri. Function)

(i) Principal Values always exit in either in Ist, IIIrd or IVth quadrant.

Quadrant for Principal Values of ITF

FUNCTION	INCLUDING QUADRANT
$\sin^{-1}$	IV th & I st
$\cos^{-1}$	I st & II nd
$\tan^{-1}$	IV th & I st

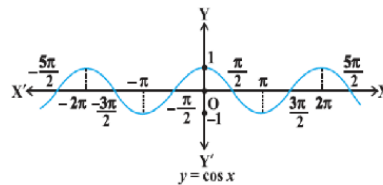
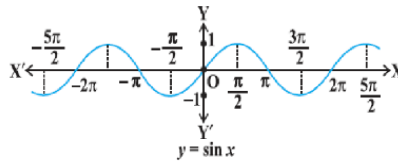
- (ii) Check the value in ITF is +ve or -ve in which quadrant?
- (iii) Now check Principal value of that ITF is lying in which two quadrant?
- (iv) Now write angle of the common quadrant

Angle in the Quadrant

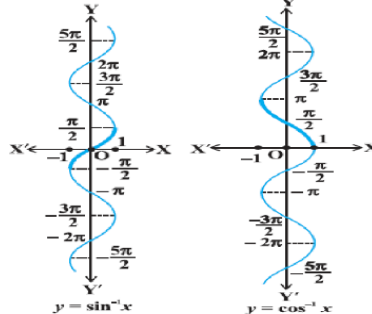
- In Ist quadrant : θ
- In IInd quadrant : $\pi - \theta$
- In IIIrd quadrant : $-\theta$

"INVERSE TRIGONOMETRIC FUNCTION"

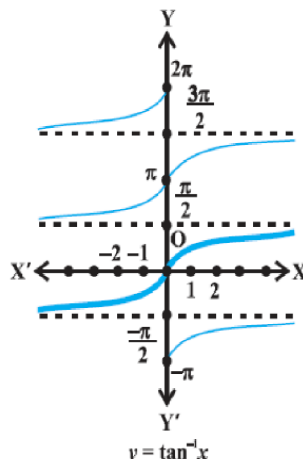
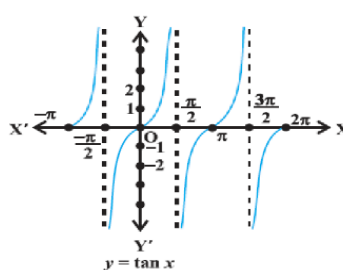
Graph of $y = \sin x$ and $y = \cos x$



Graph of $y = \sin^{-1} x$ and $y = \cos^{-1} x$



Graph of $y = \tan x$ and $y = \tan^{-1} x$



Properties of Inverse Trigonometric Function

PROPERTY - 1:

- $\sin^{-1}(\sin \theta) = \theta, \forall \theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$
- $\cos^{-1}(\cos \theta) = \theta, \forall \theta \in [0, \pi]$
- $\tan^{-1}(\tan \theta) = \theta, \forall \theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$
- $\operatorname{cosec}^{-1}(\operatorname{cosec} \theta) = \theta, \forall \theta \in [-\frac{\pi}{2}, \frac{\pi}{2}] - \{0\}$
- $\sec^{-1}(\sec \theta) = \theta, \forall \theta \in [0, \pi] - \{\frac{\pi}{2}\}$
- $\cot^{-1}(\cot \theta) = \theta, \forall \theta \in (0, \pi)$

Example

PROPERTY - 2:

- $\sin(\sin^{-1} x) = x, \forall x \in [-1, 1]$
- $\cos(\cos^{-1} x) = x, \forall x \in [-1, 1]$
- $\tan(\tan^{-1} x) = x, \forall x \in R$
- $\operatorname{cosec}(\operatorname{cosec}^{-1} x) = x, \forall x \in R - (-1, 1)$
- $\sec(\sec^{-1} x) = x, \forall x \in R - (-1, 1)$
- $\cot(\cot^{-1} x) = x, \forall x \in R$

Examples:

1. The Principal value of $\sin^{-1} \frac{1}{2} = \frac{\pi}{6}$

Hints:

- (i) Since given value is $\frac{1}{2}$ which is +ve & value of sin function is +ve in Ist quadrant.
- (ii) quadrant for $\sin^{-1}$ is IVth and Ist quadrant.
- (iii) so Principal value lie in Ist quadrant.
- (iv) $\sin^{-1} \frac{1}{2} = \frac{\pi}{6}$

2. The Principal value of $\tan^{-1}(-1) = -\frac{\pi}{4}$

Hints:

- (i) -ve of tan lie in IInd & IVth quadrant
- (ii) Quadrant of $\tan^{-1}$ is IVth & Ist quadrant
- (iii) so Principal value lie in IVth quadrant
- (iv) $\tan^{-1}(-1) = -\frac{\pi}{4}$

3. The principal value of $\cos^{-1}(-\frac{1}{2}) = \frac{2\pi}{3}$
 $\cos^{-1}(\cos(\pi - \frac{\pi}{3})) = \cos^{-1}(\cos \frac{2\pi}{3}) = \frac{2\pi}{3}$

Inverse Trigonometric Functions

1. Introduction

Inverse of a function f exists only if f is **one-one (injective)** and **onto (surjective)**. Trigonometric functions like $\sin x$, $\cos x$ etc., are not one-one over their natural domains (all reals). Hence, to define their inverses, we restrict the domain to suitable intervals where the function becomes bijective. The resulting functions are called **inverse trigonometric functions** and are widely used in calculus, engineering, and science.

2. Basic Concepts & Need for Restrictions

- If a function $f : X \rightarrow Y$ is bijective, then its inverse $f^{-1} : Y \rightarrow X$ exists. For trigonometric functions:
- Sine function** $\sin : \mathbb{R} \rightarrow [-1, 1]$ is not one-one. But if we restrict the domain to $[-\frac{\pi}{2}, \frac{\pi}{2}]$, it becomes one-one and onto $[-1, 1]$. This interval gives the **principal value branch** of $\sin^{-1}$.
 - Cosine function** becomes bijective on $[0, \pi] \rightarrow$ principal branch for $\cos^{-1}$.
 - Tangent function** on $(-\frac{\pi}{2}, \frac{\pi}{2}) \rightarrow$ principal branch for $\tan^{-1}$.
 - Cosecant, secant, cotangent** have similar restricted intervals (avoiding points where undefined).

Notation warning

$\sin^{-1} x$ **does not mean** $(\sin x)^{-1} = \frac{1}{\sin x}$. It denotes the inverse function (arc sine). The reciprocal of sine is $\csc x$.

3. Principal Value Branches (Summary Table)

The table below gives the **principal value branches** – the standard intervals chosen for each inverse trigonometric function along with their domain and range.

Function	Domain	Range (Principal Value Branch)
$y = \sin^{-1} x$	$[-1, 1]$	$[-\frac{\pi}{2}, \frac{\pi}{2}]$
$y = \cos^{-1} x$	$[-1, 1]$	$[0, \pi]$
$y = \tan^{-1} x$	$\mathbb{R}$	$(-\frac{\pi}{2}, \frac{\pi}{2})$
$y = \cot^{-1} x$	$\mathbb{R}$	$(0, \pi)$
$y = \sec^{-1} x$	$\mathbb{R} - (-1, 1)$ i.e., $ x \geq 1$	$[0, \pi] - \{\frac{\pi}{2}\}$
$y = \csc^{-1} x$	$\mathbb{R} - (-1, 1)$ i.e., $ x \geq 1$	$[-\frac{\pi}{2}, \frac{\pi}{2}] - \{0\}$

Remarks: The principal value of an inverse trigonometric function is the value lying in its respective principal branch. For example, principal value of $\sin^{-1}(1/2)$ is $\pi/6$, not $5\pi/6$.

4. Fundamental Properties & Identities

$\sin(\sin^{-1} x) = x, \ x \in [-1, 1]$	$\sin^{-1}(\sin x) = x, \ x \in [-\pi/2, \pi/2]$
$\cos(\cos^{-1} x) = x, \ x \in [-1, 1]$	$\cos^{-1}(\cos x) = x, \ x \in [0, \pi]$
$\tan(\tan^{-1} x) = x, \ x \in \mathbb{R}$	$\tan^{-1}(\tan x) = x, \ x \in (-\pi/2, \pi/2)$
$\sin^{-1}(-x) = -\sin^{-1} x$	$\cos^{-1}(-x) = \pi - \cos^{-1} x$
$\tan^{-1}(-x) = -\tan^{-1} x$	$\cot^{-1}(-x) = \pi - \cot^{-1} x$
$\sec^{-1}(-x) = \pi - \sec^{-1} x$	$\csc^{-1}(-x) = -\csc^{-1} x$
$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}, \ x \in [-1, 1]$	$\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}, \ x \in \mathbb{R}$
$\sec^{-1} x + \csc^{-1} x = \frac{\pi}{2}, \ x \geq 1$	

Composite & transformation formulas (from examples & exercises)

- $\sin^{-1}(2x\sqrt{1-x^2}) = 2\sin^{-1} x$ for $-\frac{1}{\sqrt{2}} \leq x \leq \frac{1}{\sqrt{2}}$.
- $\sin^{-1}(2x\sqrt{1-x^2}) = 2\cos^{-1} x$ for $\frac{1}{\sqrt{2}} \leq x \leq 1$.
- $3\sin^{-1} x = \sin^{-1}(3x - 4x^3), \ x \in [-\frac{1}{2}, \frac{1}{2}]$.
- $3\cos^{-1} x = \cos^{-1}(4x^3 - 3x), \ x \in [\frac{1}{2}, 1]$.
- $\tan^{-1}\left(\frac{\cos x}{1-\sin x}\right) = \frac{\pi}{4} + \frac{x}{2}, \ x \in \left(-\frac{3\pi}{2}, \frac{\pi}{2}\right)$.
- $\cot^{-1}\left(\frac{1}{\sqrt{x^2-1}}\right) = \sec^{-1} x, \ x > 1$.
- $\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right) = \frac{1}{2}\tan^{-1} x, \ x \neq 0$.

5. Important Equations (from miscellaneous exercise)

- $\tan^{-1}\left(\frac{1-x}{1+x}\right) = \frac{1}{2}\tan^{-1}x, x > 0.$
- $\tan^{-1}(\cos x) = \tan^{-1}(2 \csc x) \Rightarrow \cos x = 2 \csc x$ etc.
- $\sin^{-1}(1-x) - 2 \sin^{-1}x = \frac{\pi}{2} \Rightarrow x = 0$ or $x = 1/2$ (check domain).

6. Summary of Key Points

- Inverse trigonometric functions are defined after restricting the original trigonometric functions to specific intervals (principal branches).
- Each inverse trigonometric function has a well-defined domain and range (see table).
- $\sin^{-1}x, \tan^{-1}x, \csc^{-1}x$ are odd functions; $\cos^{-1}x$ and $\cot^{-1}x$ are neither odd nor even.
- Composition identities: $\sin(\sin^{-1}x) = x$, but $\sin^{-1}(\sin x) = x$ only when x is in principal branch range.
- The principal value must always be taken from the standard intervals given.
- Inverse trigonometric functions are essential for integration, solving trigonometric equations, and many applied problems.

✦ Historical Note (Brief)

The development of trigonometry began in ancient India. Aryabhata (476 A.D.), Bhaskara I & II contributed significantly. The notation $\sin^{-1}x$ was introduced by Sir John F.W. Herschel in 1813. Indian mathematicians also contributed to the expansion formulas and sine values for specific angles.